WE CODE CODERS HACKATHON — ROUND 1 SUBMISSION BLUEPRINT (THE PLAN)

VeriLoom AI: Intelligent Brand-Governance & Explainable Compliance Platform

Pre-Publication Multimodal Content Auditing, Citation-Grounded Remediation, and Role-Based Governance

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1. UNDERSTANDING OF THE GIVEN PROBLEM STATEMENT

1.1 Core Problem Analysis & Real-World Failure Modes

Modern growing organisations publish high volumes of marketing, sales, and corporate communications through distributed employees, external agencies, and generative AI tools. While content creation has become decentralised and instantaneous, brand governance remains manual and fragmented. As a result, four high-risk failure modes frequently reach production before anyone notices:

- **Unsupported Promotional & Factual Claims:** Creators and generative AI tools inadvertently introduce outdated pricing, unverified performance statistics, or exaggerated comparative claims that erode customer trust and expose the organisation to regulatory liability.
- **Inconsistent Tone, Voice & Terminology:** Distributed contributors use conflicting brand voices, deprecated product names, or unapproved industry terminology across different publishing channels.
- **Off-Brand Visual Identity:** Graphics, banners, and documents are published with unapproved colour hex codes, improper logo placement, or inconsistent typographic hierarchy.
- **Accessibility & Readability Deficits:** Visual assets and formatted documents fail WCAG contrast-ratio standards, lack descriptive alt-text, or use illegible text overlays.

1.2 Evidence Showing Why Existing Workflows Fail

Organisational research and workflow observations reveal three structural gaps in current brand review processes: (1) **The Static PDF Bottleneck**—brand guidelines and approved fact sheets live in 60-page PDF manuals that creators rarely cross-reference under tight deadlines; (2) **Binary Rejection Friction**—human reviewers act as late-stage gatekeepers who either miss subtle errors due to review fatigue or reject drafts without actionable fixes, forcing slow rewrite cycles; and (3) **Shadow AI Confidentiality Risks**—employees paste unreleased product copy into public AI tools for editing, risking exposure of confidential brand materials.

1.3 Specific Part of the Problem Solved by Team opopopxxnot

Team **opopopxxnot** addresses the **Pre-Publication Multimodal Audit, Explainable Remediation, and Collaborative Sign-Off Gap**. Rather than acting as an opaque blocker that automatically overwrites human creativity, **VeriLoom AI** functions as an explainable brand-governance co-pilot. It ingests approved brand guidelines, verified facts, terminology glossaries, and visual assets into a protected vault; checks text, images, and PDF documents prior to publication; explains *why* each issue was flagged by citing the exact source rule; suggests minimal corrections that preserve the creator's original purpose; orchestrates a role-based review and approval workflow; and surfaces governance insights on recurring mistakes and unclear guideline clauses.

2. THE USERS AFFECTED BY THE PROBLEM

VeriLoom AI is designed for multi-stakeholder publishing environments where internal teams and external partners collaborate on public-facing assets. The table below maps each affected user group, their organisational context, how the problem impacts them, and how our platform meets their needs:

Target User Group	Organisational Setting	How the Problem Affects Them	Core Needs Delivered by VeriLoom AI
Content Creators, Copywriters & Agency Partners	Marketing teams, social media specialists, and external creative agencies drafting daily campaigns.	Searching static brand PDFs and fact sheets slows output; late-stage rejections force frustrating rewrites that dilute their creative hook.	Instant pre-flight checks with clear rule citations and one-click rewrite suggestions that preserve their original intent and voice.
Brand Managers & Legal/Comms Reviewers	Authorised reviewers responsible for verifying claims and granting final publication approval.	Manually auditing dozens of weekly drafts for factual accuracy, disclaimers, and brand voice leads to bottlenecked launches and missed errors.	Automated pre-screening that highlights exact violations, cites approved sources, tracks revision history, and records formal sign-off.
Graphic Designers & Accessibility Leads	Design studios and digital publishing teams producing banners, decks, and PDF collateral.	Subtle hex-code deviations, low foreground-background contrast ratios, and missing accessibility tags slip into live campaigns.	Automated colour-palette distance math, WCAG contrast verification, and coordinate-level visual feedback before export.
Brand Governance & Operations Leaders	Leadership overseeing brand trust, guideline maintenance, and enterprise data security.	Lack visibility into which rules teams repeatedly misunderstand or whether confidential drafts are being sent to unapproved external AI tools.	Governance analytics identifying recurring mistakes and ambiguous guidelines, paired with local PII redaction and AI usage disclosure logs.

3. PROPOSED SOLUTION & ORIGINALITY OF APPROACH

3.1 Proposed Solution: VeriLoom AI

VeriLoom AI is an end-to-end intelligent brand-governance platform built around a single source of truth called the **Living Brand Vault**. Before any text post, promotional graphic, or document goes live, VeriLoom AI audits the asset across six dimensions: (1) Tone & Voice, (2) Terminology & Lexicon, (3) Messaging & Factual/Promotional Claims, (4) Visual Palette & Asset Consistency, (5) Accessibility Compliance, and (6) Confidentiality & AI Disclosure.

Instead of simply rejecting non-compliant drafts or silently rewriting them, VeriLoom AI opens an interactive **Explainable Remediation Studio**. Each flagged span or image region displays the exact violated clause from the Brand Vault, explains the compliance risk in plain language, and offers surgical correction options that preserve the creator's communicative intent. Creators can accept a suggestion, edit manually, or invoke a tracked **Human Override with Justification**—ensuring automation supports rather than overrides human judgment.

3.2 Originality of Approach: What Makes Team opopopxxnot Different

Because every participating team receives the same brief, most solutions default to wrapping a single LLM prompt around uploaded text—an approach that hallucinates non-existent brand rules, fails mathematical visual/accessibility checks, and rewrites creative copy into robotic prose. Our architecture differentiates itself through four technical innovations:

1. **Neuro-Symbolic Hybrid Audit Engine (Deterministic Math + Citation-Constrained RAG):** We separate deterministic checks from semantic reasoning. Terminology compliance, colour-palette accuracy (CIELAB ΔE_{OO} perceptual distance), and WCAG 2.1 contrast ratios are computed via exact mathematical and lexical algorithms—never guessed by an LLM. Meanwhile, factual claims and tone are audited via Citation-Constrained Retrieval-Augmented Generation (RAG) where any flag lacking a verified database `clause_id` is programmatically suppressed.
2. **Semantic-Anchored "Intent-Preserving" Remediation:** Before proposing a text fix, our pipeline extracts the creator's communicative intent vector (call-to-action urgency, emotional register, sentence rhythm) and constrains the rewrite to modify only the non-compliant claim or term while keeping the creator's persuasive structure intact.
3. **Bidirectional Governance ("Auditing the Guidelines, Not Just the Creator"):** When multiple creators trigger the same rule violation or authorised reviewers repeatedly override a specific AI flag, our **Guideline Ambiguity Detector** flags the brand guideline clause itself as unclear and recommends how brand managers should clarify the source documentation.
4. **Pre-Flight Confidentiality Sandbox & Transparent AI Ledger:** All uploaded drafts pass through a local entity scrubber that masks confidential project codenames, unreleased numbers, and PII before any external AI API call, and every audit generates an immutable disclosure badge detailing all AI models and external services used.

4. THE COMPLETE PRODUCT WORKFLOW

The platform executes a clear six-stage workflow designed for seamless handoffs between Creators, Managers, and Authorised Reviewers:

Stage	Actor(s)	System Action & User Experience	Output Artifact
1. Brand Vault Onboarding	Brand Manager	Imports or defines brand voice rules, approved factual claims table, preferred/banned terminology glossary, hex colour palettes, and official logo assets.	Indexed Vector & Deterministic Rule Vault with unique `clause_id` tags.
2. Multi-Format Draft Submission	Creator / Agency Partner	Uploads pre-publication content (Markdown/text copy, PNG/JPEG promotional visuals, or PDF documents) and selects the target channel (e.g., LinkedIn, Web, Press).	Sanitised, PII-redacted multimodal document payload.
3. Parallel 6-Pillar Audit Engine	VeriLoom AI Engine	Simultaneously runs deterministic glossary matching, RAG fact/claim verification, tone scoring, OpenCV colour-palette distance, and WCAG 2.1 contrast checks.	Structured JSON Audit Report with inline coordinates and rule citations.
4. Explainable Remediation Studio	Creator	Reviews highlighted text spans and visual bounding boxes; inspects the cited guideline clause; applies 1-click intent-preserving fixes or logs a Human Override reason.	Remediated Revision Draft + Annotated Override Log.
5. Role-Based Review & Approval	Manager & Authorised Reviewer	Inspects side-by-side revision diffs, replies to inline comments, validates human overrides, and triggers status transition (`Changes Requested` or `Approved`).	Signed Publication Approval Record & Revision Audit Trail.
6. Governance & Ambiguity Insights	Brand Strategy Lead	Aggregates historical flags and override patterns to visualize recurring errors by category and highlight ambiguous brand guidelines needing revision.	Governance Analytics Dashboard & Guideline Improvement Alerts.

5. THE MAIN FEATURES WE INTEND TO BUILD

5.1 Core MVP Capabilities (Round 2 Baseline)

- **Living Brand Vault Manager:** Structured interface to import and manage PDF/Markdown style guides, verified fact sheets (pricing, metrics, product capabilities), terminology rules, and brand colour/logo tokens.
- **Multi-Format Ingestion Pipeline:** Unified parser supporting raw text, rich Markdown, promotional images (PNG/JPG via OCR + vision layout segmentation), and multi-page PDF documents.
- **Citation-Grounded Claim & Tone Verifier:** Decomposes drafts into atomic factual claims, cross-checks promotional superlatives and numbers against approved facts, evaluates brand tone, and attaches the exact source guideline quote to every finding.

- **Visual & Accessibility Inspector:** Extracts dominant colours from graphics to compute perceptual distance against brand hex codes, verifies logo integrity, calculates WCAG 2.1 AA text contrast ratios, and checks alt-text presence.
- **Intent-Preserving Co-Pilot & Human Override Controls:** Side-by-side diff studio offering context-aware rewrites alongside a non-destructive "Keep Original with Justification" workflow.
- **Role-Based Collaboration & Revision Tracker:** Full state-machine supporting *Creator*, *Manager*, and *Authorised Reviewer* permissions, inline threaded comments, version history comparison, and explicit AI usage disclosure tags.

5.2 The Three Named Improvements (Planned for Round 2 → Round 3 Progression)

To directly fulfill the Round 2 and Round 3 judging criteria (*"The three improvements you named / delivered"*), we define three high-impact upgrades on top of the core workflow:

1. **Named Improvement 1 — Coordinate-Level Visual Bounding-Box Overlay & Live Contrast Simulator:** Upgrading image and PDF checks from whole-file flags to interactive canvas overlays that draw bounding boxes around off-brand visual regions and provide a real-time colour slider to preview WCAG-compliant contrast fixes.
2. **Named Improvement 2 — Guideline Blindspot & Override Intelligence Engine:** Automatically clustering reviewer override justifications and high-frequency creator violations to pinpoint confusing or outdated rules inside the Brand Vault and suggest clearer guideline phrasing.
3. **Named Improvement 3 — Local Confidentiality Redaction Sandbox & Exportable Compliance Certificate:** Adding a pre-flight regex/NER redaction toggle that strips confidential project codenames and sensitive figures before cloud inference, paired with a downloadable, tamper-evident PDF **Brand Governance Approval Certificate** upon final sign-off.

6. TECHNOLOGY STACK & TECHNICAL PLAN

6.1 Technology Stack Selection & Justification

Architecture Layer	Selected Technologies	Why This Stack Was Selected
Frontend & Workspace UI	Next.js 14 (React, TypeScript), Tailwind CSS, Shadcn UI	Enables responsive split-pane diff editors, interactive HTML5 canvas overlays for visual annotations, and clean role-based review dashboards.
Backend Orchestration	FastAPI (Python 3.11), Pydantic v2, AsyncIO	High-concurrency async execution runs deterministic rules, OpenCV vision math, and RAG LLM calls in parallel with strict JSON schema validation.
Multimodal AI & RAG Engine	Google Gemini 2.5 Flash API + Structured Outputs + Hybrid BM25/Dense Retrieval	Native multimodal comprehension (text, images, PDFs), sub-2-second latency, generous context window, and deterministic schema enforcement for citations.
Database & Vector Vault	PostgreSQL (Supabase) + `pgvector`	Unifies relational workflow data (roles, revision diffs, comments, audit logs) and vector embeddings of brand guidelines inside one transactional store.

Architecture Layer	Selected Technologies	Why This Stack Was Selected
Deterministic Vision & Accessibility	OpenCV, Pillow, Tesseract OCR, `colormath`, WCAG Luminance Engine	Executes exact CIELAB ΔE_{00} colour distance and WCAG 2.1 relative luminance calculations without relying on probabilistic LLM estimates.
Security, Privacy & Auth	NextAuth RBAC + Local Entity Redactor (Presidio / Regex AST)	Enforces Creator, Manager, and Reviewer access control and masks confidential tokens locally before external API transmission.

6.2 Most Technically Challenging Part & Architectural Solution

The Technical Challenge — Zero-Hallucination Claim Verification & Multimodal Cross-Alignment: The hardest technical challenge in automated brand governance is preventing two failure modes simultaneously: (1) *Hallucinated Compliance Flags*, where an LLM invents a brand restriction not present in the approved guidelines, and (2) *Multimodal Blindness*, where an LLM miscalculates colour hex distances or text-on-image contrast ratios, or rewrites a flagged promotional claim in a way that strips away the creator's persuasive intent.

How Our Planned 3-Stage Architecture Solves It:

1. **Stage A — Atomic Claim Decomposition & Entailment-Gated RAG:** Rather than prompting an LLM with an entire document at once, the FastAPI pipeline decomposes copy into discrete atomic statements ($C_1, C_2, ..., C_n$). Each statement queries the `pgvector` Brand Vault via hybrid search (keyword BM25 + semantic cosine similarity). The model is constrained by a Pydantic schema requiring a valid `source_clause_id` and `verbatim_source_quote`. A backend post-validator checks that the cited `source_clause_id` exists in PostgreSQL and matches the quote; if not, the ungrounded flag is discarded.
2. **Stage B — Deterministic Vision & Luminance Physics Pipeline:** For images and PDFs, OpenCV extracts dominant colour clusters and converts sRGB values into the CIELAB perceptual space to compute exact colour deviation ($\Delta E_{00} \leq 5.0$ threshold) against official brand tokens. For text regions detected via OCR bounding boxes, our engine calculates exact relative luminance $L = 0.2126R + 0.7152G + 0.0722B$ and contrast ratio $CR = (L_{lighter} + 0.05) \div (L_{darker} + 0.05)$ against WCAG 2.1 AA thresholds (4.5:1 for normal text, 3.0:1 for large text).
3. **Stage C — Intent-Constrained Minimal Edit Generator:** When generating a suggested fix, the engine passes the original span, the violated rule, and an extracted intent signature (CTA verb, tone polarity, length constraint) to produce a minimal token diff that resolves the violation while preserving the creator's original purpose.

7. RESPONSIBILITIES OF EACH TEAM MEMBER

Team **opopopxxnot** operates as a focused solo-developer unit led entirely by **Aaryan Sharma**. Operating as a solo architect eliminates inter-member communication lag and merge conflicts, allowing rapid vertical integration across all four engineering workstreams:

Team Member	Engineering Workstream	Specific Responsibilities Across All Three Rounds
Aaryan Sharma *(Solo Developer & System Architect)*	**1. Core AI, RAG & Claim Verification Engine**	Designs the FastAPI backend, PostgreSQL + `pgvector` schema, hybrid BM25/vector retrieval pipeline, atomic claim decomposition, Pydantic citation-grounding validators, and intent-preserving rewrite prompts.
Aaryan Sharma	**2. Deterministic Vision & Accessibility Pipeline**	Implements OpenCV/Pillow image and PDF parsing, OCR text-region bounding boxes, CIELAB ΔE_{OO} brand palette verification, and WCAG 2.1 relative luminance contrast calculators.
Aaryan Sharma	**3. Frontend Remediation Studio & RBAC Workflow**	Builds the Next.js 14 Brand Vault configurator, split-screen Explainable Remediation Studio, inline diff viewer, Human Override modal, and role-based approval workflow (Creator, Manager, Reviewer).
Aaryan Sharma	**4. Governance Analytics, Privacy & QA Testing**	Develops the local PII/confidentiality redaction filter, AI disclosure ledger, Guideline Ambiguity Analytics dashboard, benchmark evaluation datasets, end-to-end reliability testing, and deployment.